Explainable Machine Learning and Statistics for Early Bankruptcy Prediction: Evidence from the Taiwan Bankruptcy Dataset

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Abstract: Bankruptcy prediction is vital for stakeholders, yet most studies focus on short-term horizons, leaving little room for corrective action. This study extends the timeline by developing machine learning models to predict bankruptcy risk 2–3 years in advance using the Taiwan bankruptcy dataset. Unlike traditional approaches limited to a few financial ratios, we utilize a high-dimensional set of profitability, liquidity, leverage, efficiency, and cash flow indicators. Since the class imbalance is severe, we use resampling techniques and cost-sensitive learning. Several machine learning models were trained, including Logistic Regression, Random Forest, XGBoost, and ensemble voting classifiers with a focus on demonstrating predictive accuracy, rare event sensitivity, and confidence. Explainable AI mechanisms are also utilized to illustrate important financial ratios that contribute to early risk indicators. The study introduces enhancements to the current learning by integrating meaningful financial measures, imbalance management methodologies, and interpretable machine learning, alongside the probable usefulness of proactive and transparent early warning systems.

Keywords: Bankruptcy Prediction; Financial Distress; Early Warning Systems; Machine Learning; Imbalanced Data; Cost-Sensitive Learning; Synthetic Minority Oversampling Technique (SMOTE); Logistic Regression; Random Forest; XGBoost; Ensemble Learning; Explainable Artificial Intelligence (XAI); SHAP (Shapley Additive Explanations); Financial Ratios; Profitability; Liquidity; Leverage; Taiwan Bankruptcy Dataset; Predictive Modeling; Corporate Failure Forecasting; Financial Risk Management.

1. Introduction

Forecasting of business failure has been a major focus of financial studies because of its significance to investors, creditors, officials, and politicians. Signalling the possible bankruptcy of firms will be very essential since stakeholders will be able to intervene by arranging policies in the banks they provide loans, organising debt structure or making wise decisions on where to invest the money. Such traditional models as the Z-score created by Altman and the O-score created by Ohlson were common in research and practice. These models apply a few financial ratios and statistical procedures and are simple and clear. However, they have difficulty capturing the dynamic and intricate nature of financial distress in current businesses. With more high-dimensional financial data available and improvements in machine learning, there is a chance to create stronger frameworks for predicting business failure, well before it happens.

Despite a wealth of research on bankruptcy prediction, some limitations remain. First, many studies concentrate on post-failure or near-failure predictions, usually using data from the year just before bankruptcy. These short-term perspectives give limited warning and are useless for preventive decision-making. Second, older models do not typically rely on more than a few financial indicators, ignoring the larger picture of indicators of financial stability, such as profitability, liquidity, leverage, efficiency, and cash flow. Third, the real-world datasets rarely contain business failure, and the bankruptcy numbers are probably less than 5 percent of the total sample. Many existing studies do not effectively address this class imbalance, resulting in biased models struggling to recognize at-risk firms. Finally, while advanced machine learning techniques like Random Forests and Gradient Boosting have shown potential in other financial fields, their consistent use in early bankruptcy prediction is still limited, especially regarding incorporating interpretability mechanisms to help stakeholders understand why a firm is considered high-risk.

This study aims to fill these gaps by creating and testing machine learning models for early bankruptcy prediction, using the Taiwan Bankruptcy Dataset, which includes a wide range of financial ratios. The contributions of this

research are fourfold. First, it shifts the focus from short-term failure detection to early prediction, aiming to spot risk 2-3 years in advance. Second, it utilizes a very broad, high-dimensional set of data that captures various classes of financial ratios as opposed to a thin subset on indicators. Third, the authors directly address the severe class imbalance in the data on bankruptcies by means of resampling and cost-sensitive learning approaches, making the models more susceptible to infrequent failure occurrences. Last but not least, it also includes explainable machine learning techniques, including SHAP-based interpretation, to demonstrate the financial ratios with the strongest association with the risk of early bankruptcy. This study creates a predictive accuracy improvement and also improves the robustness and practical utility of machine learning models in financial subjugation administration.

2. Literature Review

The predictive value of bankruptcy has its roots in the statistical based models of the Z-score (Altman, 1968) and the O-score (Ohlson, 1980). These models were based on discriminant and regression on logistic respectively. They use few financial ratios so as to forecast corporate distress. They are simple to use and make assumptions about linear connections, even though the relationships in the financial markets are not always linear, but usually nonlinear. On top of this they are narrow gravitating their emphasis on few ratios -usually not more than five- which limits their overall capacity to forecast in altering business scenarios.

During the next decades, logistic Regression was the most widespread tool to make the predictions of the bankruptcy (Chaudhuri and De, 2011). Achieving this was subsequently implemented using decision trees and Support Vector Machines (SVMs) (Min and Lee 2005). Nevertheless, such models tended to be poor extrapolators particularly when the datasets are unbalanced in which cases of bankruptcies constituted a minor part of firms. Studies show that traditional methods often predict the majority non-bankrupt class accurately, but fail to recognize the minority bankrupt class (Jones et al., 2017).

Most recently, some newer learning methods such as Random Forests, Gradient Boosting Machines, and Extreme Gradient Boosting (XGBoost) are popular. They involve a set of weak learners being grouped together in order to improve predictive behaviour, stability and generalisation (Li and Sun, 2019; Barboza et al., 2017). Studies have shown that boosting methods are more successful compared to conventional models in forecasting bankruptcy with huge amounts of finance ratios. However, the majority of works are inclined to compare short-term forecasts regularly in less than a year preceding bankruptcy (Gepp, Kumar, 2021), which restricts the quality of such techniques as early warning signals.

One weakness in the prediction of bankruptcy that can hardly be resolved is the massive disproportion of bankrupt and non-bankrupt companies. Less than 5% of the total cases are usually associated with bankruptcy which means that the majority classifier will skew more towards weakened majorities.In an attempt to address it, resampling methods (including SMOTE and undersampling) and cost-sensitive learning have been employed (Brown and Mues, 2012; Bekhet and Eletter, 2014). Despite these efforts, some of these models fail to provide high recall rates particularly with bankrupt companies; this tells us that there should be a better process of coping with the imbalances that the state- of-the-art machine learning algorithms can effectively capture.

Predictive, however, accuracy is essential; interpretability is also paramount to building confidence with the stakeholders and contributing to the decision-making process. The traditional statistical models are simple to understand, whereas the complex machine learning models usually serve as black boxes. Recent studies have introduced explainable AI (XAI) techniques, including feature ranking, SHAP values and LIME explanations. These techniques assist in clarifying the monetary indicators of the model forecast (Kou et al., 2021; Lin et al., 2022). Such tools can be used to relate model predictions to certain financial indicators, such as depleting liquidity or rising leverage, and fill the divide between model, complexity, and usability.

The literature indicates that there is a clear shift in the use of simple ratio-based statistical models to more sophisticated machine learning methods. Nevertheless, there are still three significant limitations. First, the short-term focus of most models does not allow enough time for corrective action. Second, many studies depend on a

narrow range of financial indicators and overlook the multifaceted nature of bankruptcy risk. Third, although ensemble methods improve predictions, issues with class imbalance and interpretability are not properly addressed. This study aims to fill these gaps by proposing a high-dimensional, machine learning-based bankruptcy prediction model that focuses on early warnings, effective handling of imbalance, and clarity.

3. Methodology

3.1 Data Collection

The Taiwan Bankruptcy Dataset used in this study consists of financial data of 6,819 firms in a number of years. It has 95 financial ratios which serve as predictive variables. These ratios encompass five significant financial health indicators, namely, profitability, liquidity, leverage/solvency, cash flow, and efficiency. The target variable is discrete and has the value of one and zero, which shows bankruptcy and non-bankruptcy respectively. All ratios are prepared out of accounting statements prior to bankruptcy filing. This helps to make the dataset demonstrate real early-red flags not failure outcomes. One of the first issues when working with this data is the large imbalance in the classes, approximately 3 per cent of companies are marked as bankrupt.

To this end, we stratified the dataset through the sampling technique in order to preserve the imbalance between the sets of training and test. StandardScaler was used to normalize financial ratios in order to be able to compare the values of the metrics using different units. Infinite or missing values that are usually caused by division-based ratios were substituted with median values. To mitigate the imbalance further, we used the Synthetic Minority Oversampling Technique (SMOTE) to the training set, which bodes realistic synthetic bankrupt cases. This will make the models learn the bankruptcy patterns in a better way.

3.2 Model/Algorithm

Predictive or interpretability was compared on a vast set of machine learning models:

- Logistic Regression: It is grounded on the class weighting, the application of which mitigates the imbalance with the support of the class weighting as a baseline statistical model. It is applicable in benchmarking, which is its advantage of being able to be interpreted.
- Decision Tree: Has transparency and offers rule based predictions, although it overfits on imbalanced data.
- Random Forest: This is a group of trees, which increases stability on the reduction of variance, and nonlinear patterns do exist.
- XGBoost: It is a gradient boosting algorithm that has been known to handle the imbalance classification problem effectively using weighted learning.
- Support Vector Machine (SVM): It is a model that learns optimal separation boundaries; it works well in high-dimensional space, and computationally it is expensive.
- K-Nearest Neighbors (KNN): A simple similarity-based classifier; however, it is sensitive to noisy and high-dimensional data.
- Voting Classifier: A hybridization of Voting Classifiers with Logistic Regression, Random Forest, and XGBoost.

This paper gives an exhaustive evaluation of early bankruptcy prediction by subjecting the traditional statistical approaches and contemporary ensemble tools to test.

3.3 Evaluation Metrics

Given the rarity of bankruptcy events, accuracy alone is not enough and can be misleading. This work is devoted to the metrics which are applicable in the case of imbalanced classification:

- Precision The percentage of correctly predicted bankruptcies out of all the predicted bankruptcies.

- Recall (Sensitivity): This is the proportion of true bankruptcies picked up; it is given priority because it is more expensive to miss a failed firm than to precipitate a false alarm.
- F1-Score: The average of precision and recall, balancing both measures.
- Area Under the ROC Curve (AUC-ROC): Evaluates the balance between true positive and false positive rates.
- Area Under the Precision-Recall Curve (PR-AUC): Especially important for class imbalance, as it better shows performance on rare bankrupt cases.

To improve model transparency, we used SHAP (Shapley Additive Explanations) with ensemble methods. SHAP offers both global insights (which financial ratios most strongly predict bankruptcy across all firms) and local explanations(why a particular firm is likely to be in high risk). This is so that the models would not only predict bankruptcy but would also accurately give sensible signals to the stakeholders.
Because the cost of missing a bankrupt company is higher than the cost of false alarms, threshold adjustment and cost sensitive evaluation was considered a strategy to prioritize recall. Reduced classification thresholds can lead to increased sensitivity to bankrupt companies while decreasing Type II errors in financial risk prediction.

3.4 Mathematical Formulation

Logistic Regression:
The Logistic Regression model estimates the probability of bankruptcy using the sigmoid function:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n)}}$$

where:

- $Y = 1$indicates bankruptcy,
- $X_1, X_2, \ldots, X_n$represent financial ratio predictors,
- $\beta_0$is the intercept,
- and $\beta_1, \beta_2, \ldots, \beta_n$are model coefficients.

XGBoost Objective function:
XGBoost minimizes the following regularized objective function:

$$Obj = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where:

- $l(y_i, \hat{y}_i)$denotes the loss function measuring prediction error,
- and $\Omega(f_k)$represents the regularization term controlling model complexity.

Confusion Matrix:
The confusion matrix used for model evaluation is represented as:

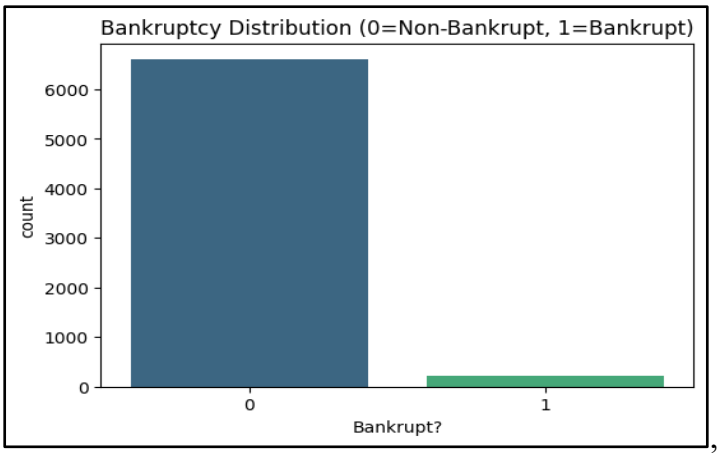
$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$$

where TP, FP, FN, and TN denote true positives, false positives, false negatives, and true negatives respectively.

4. Analysis

4.1 Dataset Imbalance

The Taiwan Bankruptcy data are highly skewed, in the sense that majority of the firms are not bankrupt. The percentage of the bankrupt companies was 3.2 out of 6,819 companies and 96.8 percent of the companies were not bankrupt. Such imbalance compares it with prediction of results since standard models tend to favor the majority class. In order to correct this issue, Synthetic Minority Oversampling Technique (SMOTE) was applied to equalize the training sample and enhance the identification of the bankrupt companies.



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Figure 1: Distribution of bankrupt vs. non-bankrupt firms in the dataset, highlighting the severe class imbalance.

4.2 Model Performances

Logistic Regression: It has the highest recall (0.7955) and thus identifies the majority of the bankruptcies yet very low precision (0.1882) implies that it produces a lot of false alarms. The overall separation power is good with good ROC-AUC (0.9170).

Random Forest: The precision is 0.4286 and the recall is 0.6136, which is better than the Logistic Regression and the accuracy is high (0.9611). It is practical because of strong discriminative ability (ROC-AUC 0.9429) although there are some bankruptcies that are missed.

Support Vector Machine (SVM): Decent recall (0.6591) and poor precision (0.1747), results in false positives occurring too often. Accuracy (0.8886) and ROC-AUC (0.9135) are no more than decent, but the imbalance reduces the reliability.

K-Nearest Neighbors (KNN): Poorest values of recall (0.6346) and Precision (0.1637), and worst ROC-AUC (0.8671). High-dimensional imbalance is sensitive, which makes it less applicable in the investment of bankruptcy.

XGBoost: Best overall model, which has the highest accuracy (0.9626) and ROC-AUC (0.9499). The values of precision (0.4386) and recall (0.5682) are satisfactory, as indicated by the optimal combination of F1-score (0.4950).

Voting Classifier: decent trade-off in the accuracy of the precision (0.3733) and the recall (0.6364). Although it is robust (ROC-AUC 0.9421), it does not perform as well as XGBoost but it is stable when used in metrics.

The models, which have the best trade-off between recall and AUC, were tree-based, which include Random Forest, XGBoost, and Voting Classifier. The Logistic Regression demonstrated excellent recall of 0.7955, but it was very low with precision of 0.1882. XGBoost had the best ROC AUC (0.9499), thus it is the best model to apply when classifying the accurate bankrupt and non-bankrupt corporations.

4.3 ROC Curve

To compare the separation performance of the models the Receiver Operating Characteristic (ROC) curves were plotted, bankrupt and non-bankrupt firms.

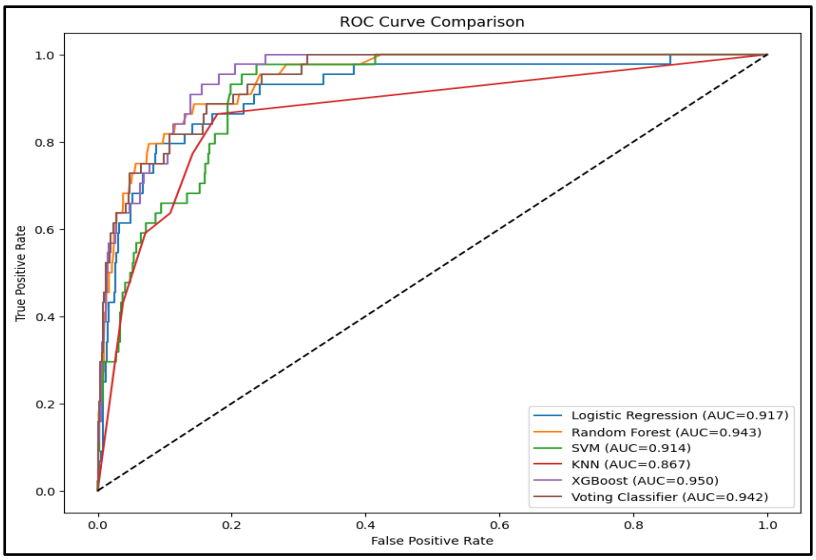


Figure 2: ROC curves for different classifiers. XGBoost achieved the highest AUC (0.9499), closely followed by Random Forest (0.9429) and Voting Classifier (0.9421).

Financial ratios based on which the models identify the early warning of the bankruptcy, can determine the decline of the financial ratios before the aforementioned time bankruptcy filing. This dataset consists of such financial ratios as Return on Assets (ROA), Debt Ratio, Current Ratio and Cash Flow Rate. Such ratios tend to give indications of trouble way before the event of bankruptcy occurs. The models can identify the minor shifts in such areas as profitability, leverage, and liquidity in different firms, which occurs prior to a corporate failure chronicling the trends within different companies. This gives stakeholders useful early warnings for taking preventive action.

4.4 Precision-Recall Curve

Besides the ROC curves, we also drew Precision-Recall (PR) curves in order to clearly evaluate the performance when there is high imbalance in the classes. ROC curves may sometimes over-emphasize performance in imbalanced sets but PR curves emphasize the ability of the model to discriminate correctly on the minority class.

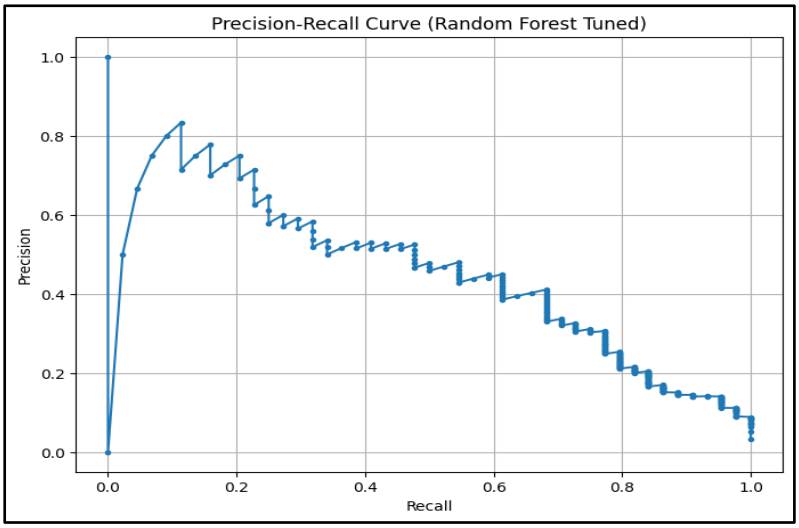


Figure 3: Precision-Recall curves of the classifiers. XGBoost and Random Forest had the greatest PR-AUC scores, which proved their greater capacity to identify the rare cases of bankruptcy as compared to the Logistic Regression and KNN.

4.5 SHAP Analysis

To understand the predictions made by the machine learning models, we used the SHapley Additive exPlanations (SHAP) framework. SHAP provides a global perspective of the importance of features as it provides the importance of factors that seem to be of the greatest importance across all the firms and local explanations, that is, explanations of how a certain firm was or was not predicted to be bankrupt. This renders the models precise and clear which is very crucial in the generation of financial decisions.

The SHAP summary plot reveals the main characteristics which influence the predictions of bankruptcy. Predictors The largest predictors were ratios involving leverage (debts ratio), liquidity (current ratio and working capital/ total assets), profitability (rate of return on assets and cash flow). The model indicated that firms with declining profitability and higher leverage faced a greater risk of bankruptcy.

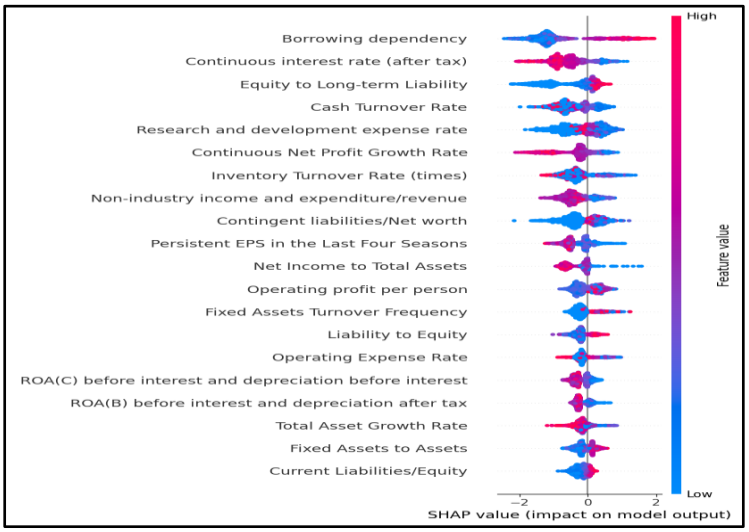


Figure 4: SHAP summary plot showing top predictors of bankruptcy. Leverage and profitability ratios are the strongest contributors to bankruptcy risk.

In addition to global interpretation, SHAP force plots were used to explain individual firm predictions. These visualizations show how specific financial ratios influence a prediction toward bankruptcy (red) or non-bankruptcy (blue). For instance, a firm with a high debt ratio and negative cash flow was classified as bankrupt. In contrast, strong current ratios and profitability indicators lowered bankruptcy risk for other firms.

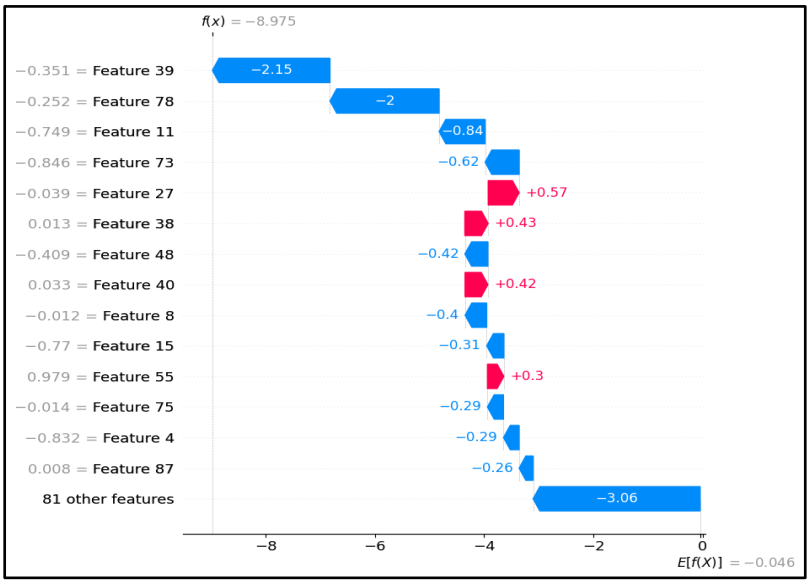


Figure 5: SHAP force plot for an individual firm prediction. Red features push the model toward predicting bankruptcy, while blue features mitigate risk.

The research comprised of SHAP led to high predictive performance and useful insights of the conduct of the study. This interpretability enables the lenders, investors and regulators to put their trust on the models in pointing out the most essential financial indicators in the display of early bankruptcy risk. For instance, companies that were classified as high-risk generally had high debt ratios, poor cash flow and declining profitability. In such instances, managers might scale back on leverage and enhance liquidity administration, lenders might string up credit monitoring, and regulators could ramp up scrutiny of badly hit sectors and industries. This interpretation of the framework in the context of cases provides a more practical interpretation for actual decision making.

4.6 Practical Implications

The suggested framework is of immense importance to practitioners. To the lender, it is much easier to find the firms in danger early to cover credit risks in advance and reorganise rather than deal with expensive banks. The regulators can use the model as a tool whereby the weak sectors or regions are identified in the early stages. The corporate managers get to learn also which financial indicators are best at indicating the distress such that they can undertake corrective measures in time regarding leverage, liquidity or profitability policies. The framework is able to predict bankruptcy in addition to considering the interpretation of the methodology by incorporating the use of SHAP thus, signaling financial levers that could be used to mitigate risk exposure.

5. Discussion

The results provide sharp contrasts between the models in dealing with the situation when it is not suitable on the Taiwan Bankruptcy dataset since the imbalance of classes is high. The Logistic Regression had the highest recall (0.7955), therefore it was useful in detecting the bankruptcies, though its very low precision resulted in large false positives. KNN and SVM fared fairly well, exhibiting inabilities to function in high-dimensional, imbalanced environments. Consistently, tree-based models performed better compared to others, and Random Forest and the Voting Classifier record a good balance between recall, precision, and ROC-AUC. XGBoost proved to be the most successful algorithm and provided the best possible results, including the highest accuracy (0.9626) and ROC-AUC (0.9499), and median F1-score (0.4950). The performance of XGBoost was once again reaffirmed using precision-recall analysis, showing that XGBoost has a better performance in finding rare cases of bankruptcy. The leverage and profitability ratios were the most important predictors indicated by the transparency of SHAP analysis. Therefore, though the usual realities of previously ensured stability by ensemble approaches hold true, XGBoost can be viewed as the most trusted and most consistent predictor of early bankruptcy, which can be interpreted as well as provide a high performance with a high degree of trust by the stakeholders.The results indicate that threshold tuning and cost sensitive optimization can further enhance recall, meaning that fewer high-risk businesses will be missed. The results are aligned with previous research that also reveals the better performance of ensemble learning methods when facing imbalanced sets of bankruptcies. The power of XGBoost highlights the ability of boosting algorithms to better understand the financial distress patterns as they tend to be more nonlinear in nature.

Table 1: Performance comparison of machine learning models for bankruptcy prediction.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC-AUC**
Logistic Regression	0.8827	0.1882	0.7955	0.3043	0.9170

Random Forest	0.9611	0.4286	0.6136	0.5047	0.9429
SVM	0.8886	0.1747	0.6591	0.2762	0.9135
KNN	0.8834	0.1637	0.6346	0.2065	0.8671
XGBoost	0.9626	0.4386	0.5682	0.4950	0.9499
Voting Classifier	0.9538	0.3733	0.6364	0.4706	0.9421

5.1 Limitations

Although there were successful outcomes, there were some limitations to this study. First, the analysis is based on Taiwan Bankruptcy Dataset, which may not be generalizable to other industrial sectors or other countries. Second, the study primarily relies on financial ratios and does not include macroeconomic, governance or textual indicators. Third, while SHAP makes the interpretability easier, complex ensemble models can still be challenging for non-technical stakeholders to comprehend. Future research is needed to overcome these shortcomings, in particular with larger multi-country data sets and hybrid explainable approaches.

6. Conclusion

The analysis contributed to the research in the field of predicting bankruptcy by extending the traditional models of prediction on a short synoptical scale and elaborated on machine learning frameworks that would be able to predict the risk 2-3 years in advance. Using the Taiwan Bankruptcy Dataset, which is highly imbalanced with only 3.2% bankrupt firms, the study applied SMOTE resampling and cost-sensitive learning to improve the identification of uncommon failure cases. Various models were experimented and it was found that tree-based models, more so XGBoost came out with the best compromise in accuracy, recall, precision and ROC-AUC, thus making it the most reliable and robust. Importantly, the integration of SHAP-based interpretability allowed stakeholders to understand the role of key financial ratios, such as leverage and profitability indicators in making projections. This study will equip investors, regulators and managers with a transparent and efficient early-warning framework, by applying high-dimensional financial indicators, imbalance-handling strategies as well as explainable AI that can aid in managing financial risks proactively.

Future Scope

While the current work shows promising results, several directions still await further studies. Research in the future can also be devoted to the further enlargement of the dataset by adding, say, various industries, more extended time frames, and data stored in different sources to enhance the generalization of the obtained model. Also, the application of deep learning architectures, ensemble, and explainable AI could be used to improve the performance and simplify the results. Last but not the least, it will be necessary to implement the work practically and prove it on a partnership basis with financial institutions, healthcare providers, and government agencies, which will help to make the work more practical and significant. Hybrid models, which integrate interpretable statistical models (e.g., Logistic Regression) with powerful ensemble methods (e.g., XGBoost or

Random Forest) could be investigated in future research. These hybrid techniques may offer a more balanced blend of predictive power and model interpretability, enhancing trust and usability in financial risk assessment for stakeholders.

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